11. Simulation and Performance Analysis of Big Data Processing for IoE Sensor Data using Hadoop and Apache Spark

Objective

1.To understand the working principles of Hadoop MapReduce and Apache Spark.

2.To simulate large-scale IoE sensor data processing using Python.

3.To calculate the execution time and throughput of both processing frameworks.

4.To compare the performance of Hadoop MapReduce and Apache Spark.

5.To visualize the performance comparison using graphs.

code

```
import matplotlib.pyplot as plt


# ====================================
# HADOOP vs APACHE SPARK COMPARISON
# ====================================

# Number of IoE sensor records
records = [100000, 500000, 1000000, 5000000, 10000000]

# Assumed execution time (seconds)
hadoop_time = [8, 25, 45, 180, 350]
spark_time = [5, 14, 25, 90, 170]

# ====================================
# PERFORMANCE CALCULATIONS
# ====================================

# Throughput = Records / Execution Time
hadoop_throughput = [
```

```
    n / t for n, t in zip(records, hadoop_time)
]

spark_throughput = [
    n / t for n, t in zip(records, spark_time)
]

# Speedup = Hadoop Time / Spark Time
speedup = [
    h / s for h, s in zip(hadoop_time, spark_time)
]

# Improvement percentage
improvement = [
    ((h - s) / h) * 100
    for h, s in zip(hadoop_time, spark_time)
]

# ====================================
# DISPLAY RESULTS
# ====================================

print("HADOOP MAPREDUCE vs APACHE SPARK")
print("-" * 85)

print("Records\t\tHadoop(s)\tSpark(s)\tHadoop TP\tSpark TP")
print("-" * 85)
```

```
for i in range(len(records)):
    print(
        f"{records[i]:,}\t"
        f"{hadoop_time[i]}\t\t"
        f"{spark_time[i]}\t\t"
        f"{hadoop_throughput[i]:.2f}\t"
        f"{spark_throughput[i]:.2f}"
    )

print("\nSpeedup:", [round(x, 2) for x in speedup])
print("Improvement (%):", [round(x, 2) for x in improvement])

# ====================================
# GRAPH 1: EXECUTION TIME
# ====================================

plt.figure(figsize=(9, 5))

plt.plot(
    records, hadoop_time,
    marker='o', linewidth=2,
    label='Hadoop MapReduce'
)

plt.plot(
    records, spark_time,
    marker='s', linewidth=2,
    label='Apache Spark'
```

```
)

plt.xlabel("Number of IoE Sensor Records")
plt.ylabel("Execution Time (Seconds)")
plt.title("Hadoop vs Spark - Execution Time Comparison")

plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

# ====================================
# GRAPH 2: THROUGHPUT
# ====================================

plt.figure(figsize=(9, 5))

plt.plot(
    records, hadoop_throughput,
    marker='o', linewidth=2,
    label='Hadoop MapReduce'
)

plt.plot(
    records, spark_throughput,
    marker='s', linewidth=2,
    label='Apache Spark'
)
```

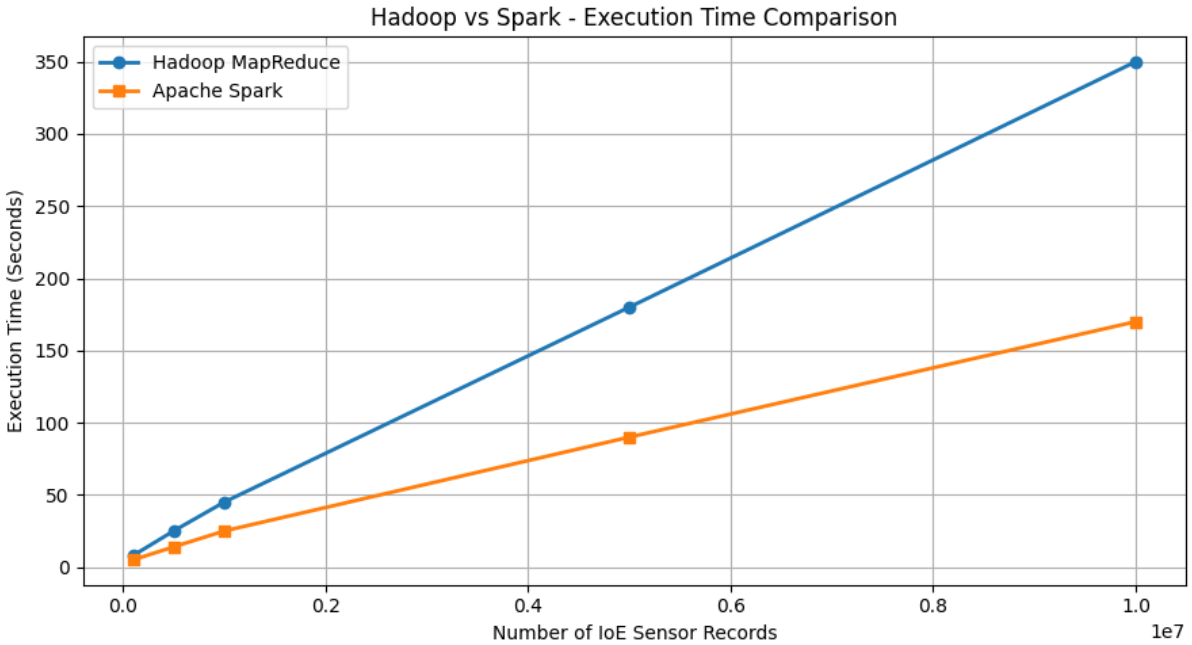
```
plt.xlabel("Number of IoE Sensor Records")
plt.ylabel("Throughput (Records/Second)")
plt.title("Hadoop vs Spark - Throughput Comparison")

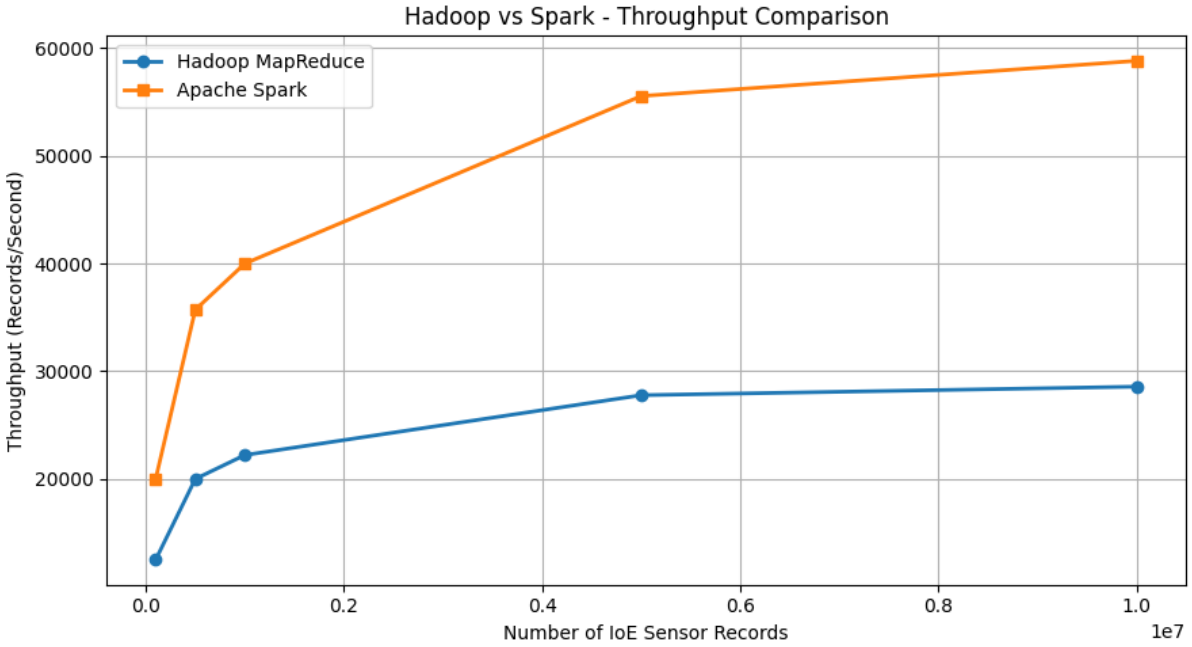
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

HADOOP MAPREDUCE vs APACHE SPARK				
Records	Hadoop(s)	Spark(s)	Hadoop TP	Spark TP
100,000	8	5	12500.00	20000.00
500,000	25	14	20000.00	35714.29
1,000,000	45	25	22222.22	40000.00
5,000,000	180	90	27777.78	55555.56
10,000,000	350	170	28571.43	58823.53

Speedup: [1.6, 1.79, 1.8, 2.0, 2.06]

Improvement (%): [37.5, 44.0, 44.44, 50.0, 51.43]


Hadoop vs Spark - Execution Time Comparison
Hadoop MapReduce
Apache Spark
Execution Time (Seconds)
0
50
100
150
200
250
300
350
0.0
0.2
0.4
0.6
0.8
1.0
1e7
Number of IoE Sensor Records


Hadoop vs Spark - Throughput Comparison
Hadoop MapReduce
Apache Spark
Throughput (Records/Second)
20000
30000
40000
50000
60000
0.0
0.2
0.4
0.6
0.8
1.0
1e7
Number of IoE Sensor Records